PHYSMOD Lyon 2024

La Région Auvergne-Rhône-Alpes · ÉCOLE CENTRALE LYON · Lyon 1 · INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON · cnrs · LMFA

Interaction between the large scales in the boundary layer and street canyon flow

Authors:

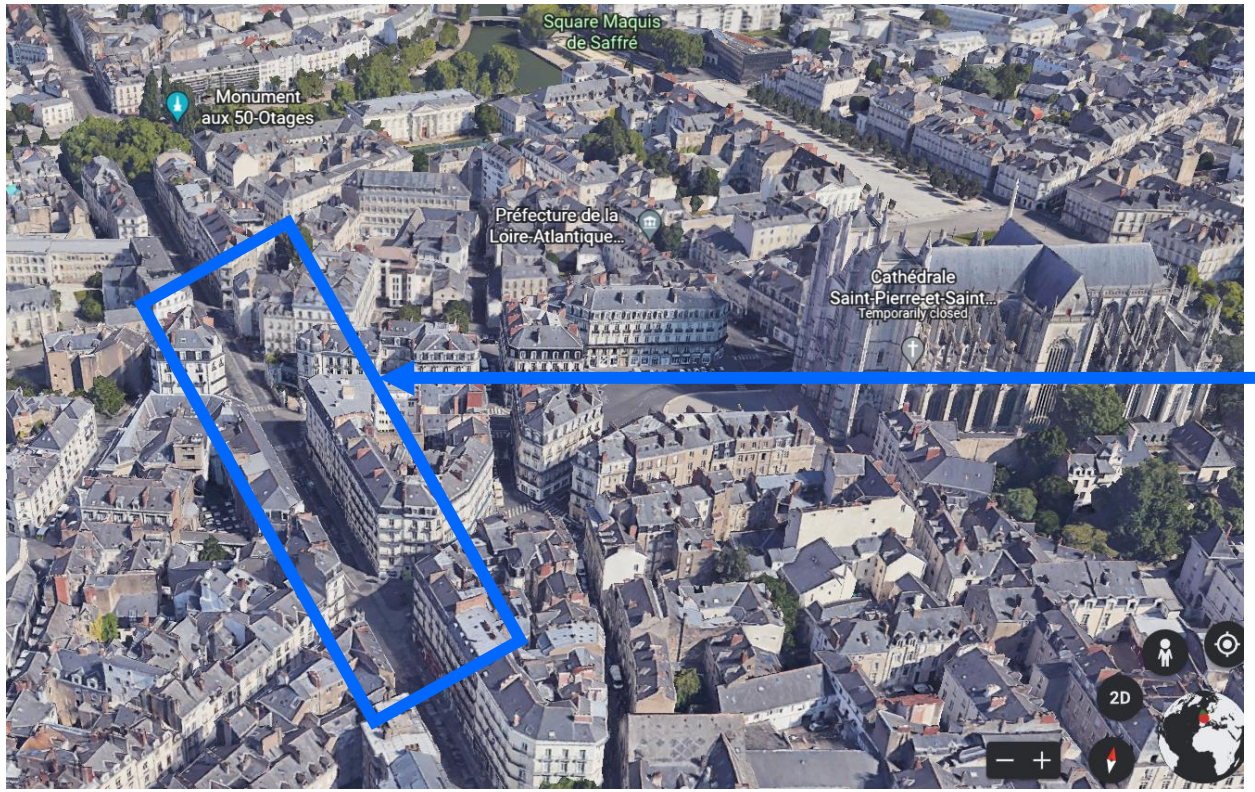
Haoran Du 

Laurent Perret 

Eric Savory 

Background

> Background



Real Satellite Google Map-Rue de Strasbourg (RdS)

Poor urban air quality

↓

Dynamic transport of air and pollutants

↓

Street canyon model ← Urban boundary layer

↓

Interaction

↓

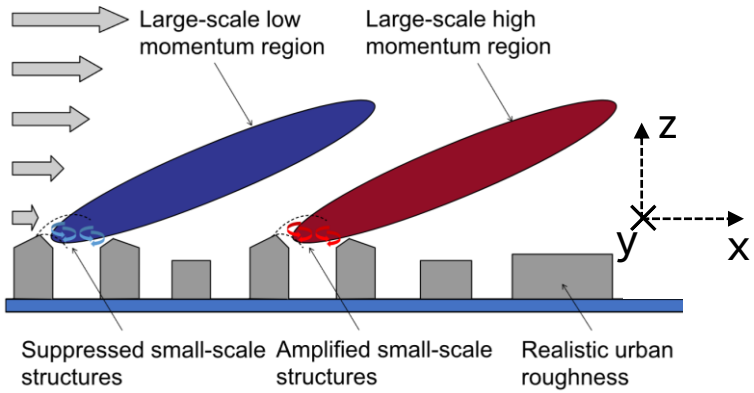
GOAL

Urban-scale dynamical model

Research Questions

Motivation

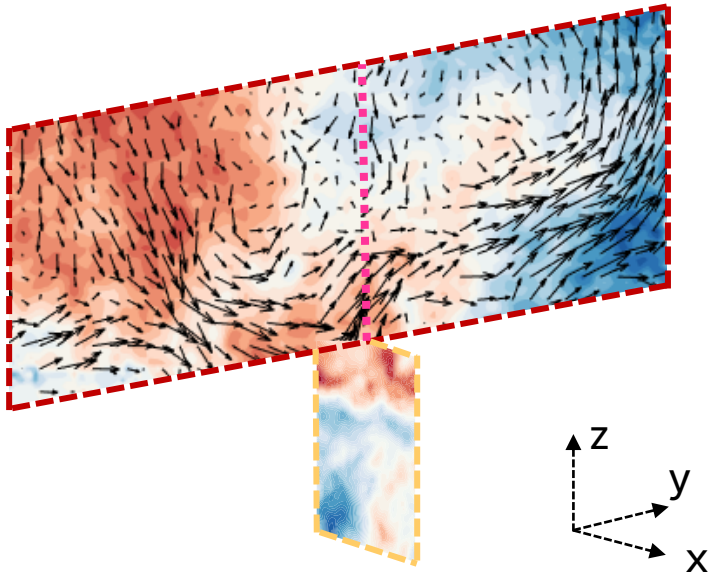
Previous work showed non-linear interaction between the large scales in boundary layer (streamwise-vertical direction) and the small scales in the street canyon.



(Du et al., BLM, 2024)

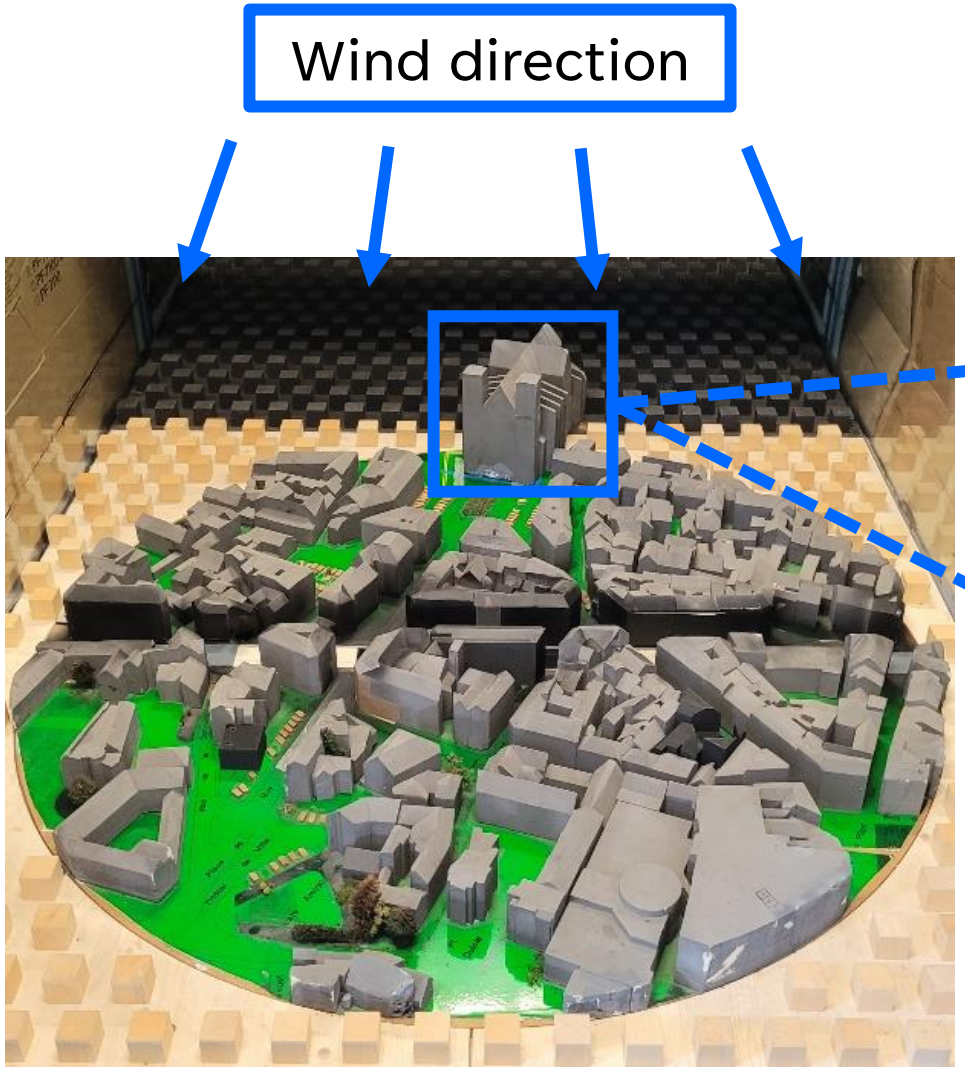
Questions

1. How is boundary layer (large-scale) in longitudinal-vertical direction like at the same time when we have street canyon flow measured in streamwise-vertical direction?
2. How do the large scales interact or trigger the flow in street canyon?



Experimental Set-ups

> Experimental Set-ups



1:200 scaled model – (RdS)

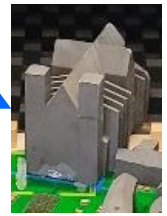
** $\bar{h} = 2h_1 = 100\ mm = 0.86h_2$



$B1\bar{h}$: 1-block

$2\bar{h} \times 2\bar{h} \times \mathbf{1}\bar{h}$

90° to the canyon



$Cath$: Cathedral

$2\bar{h} \times \mathbf{3.5}\bar{h} \times \mathbf{3}\bar{h}$

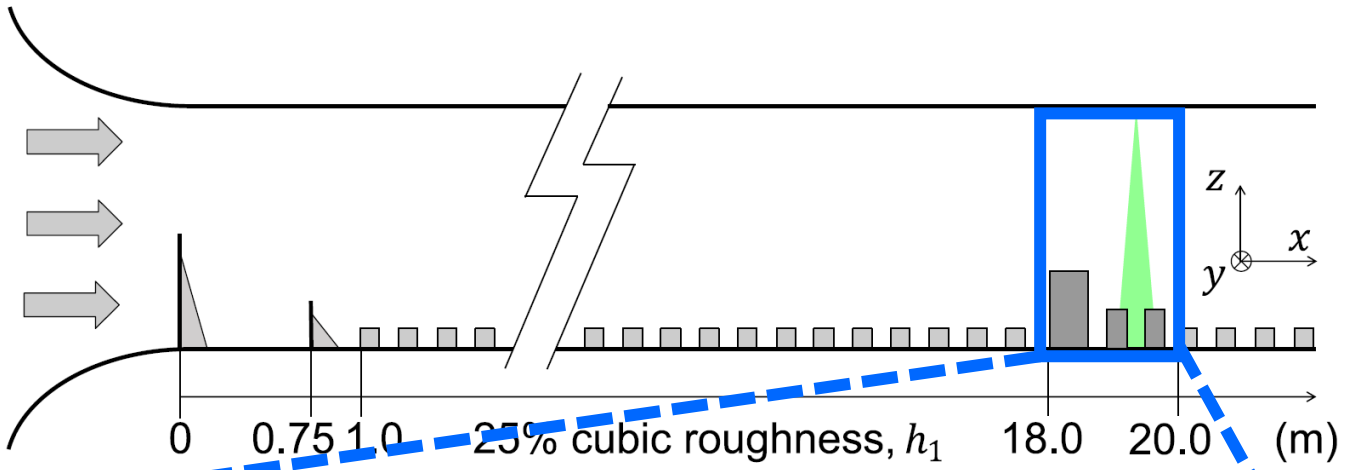
~70° to the canyon

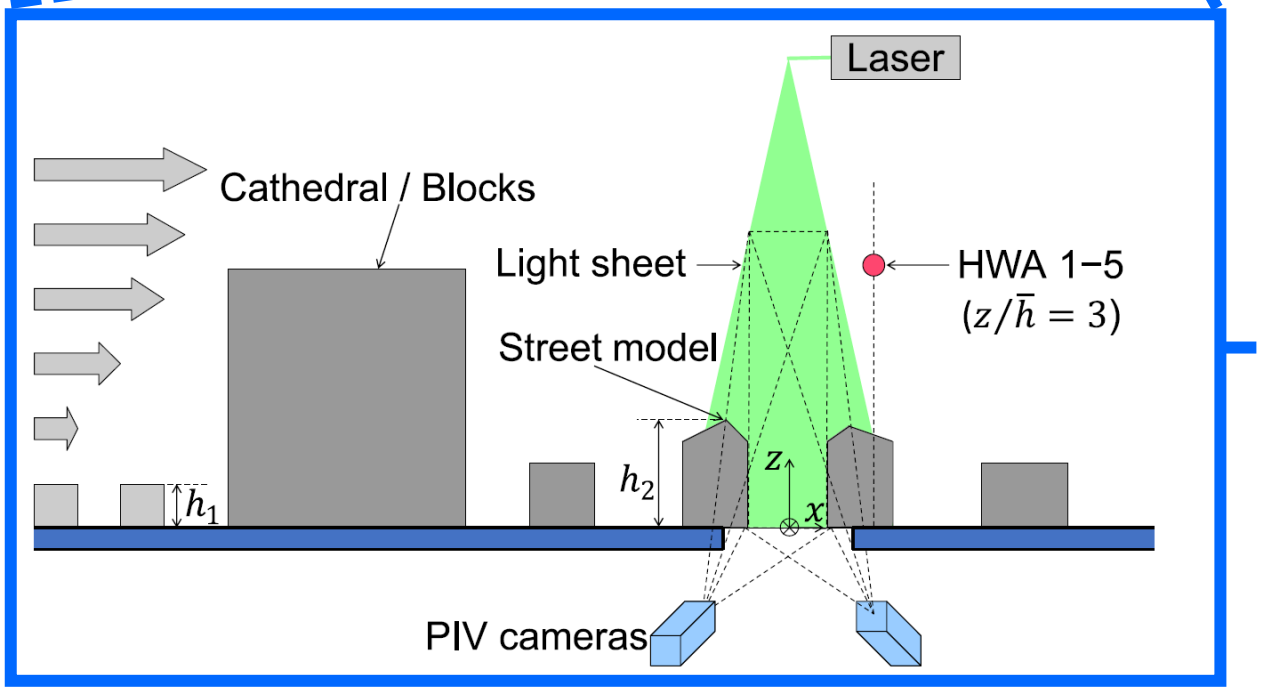
> Experimental Set-ups

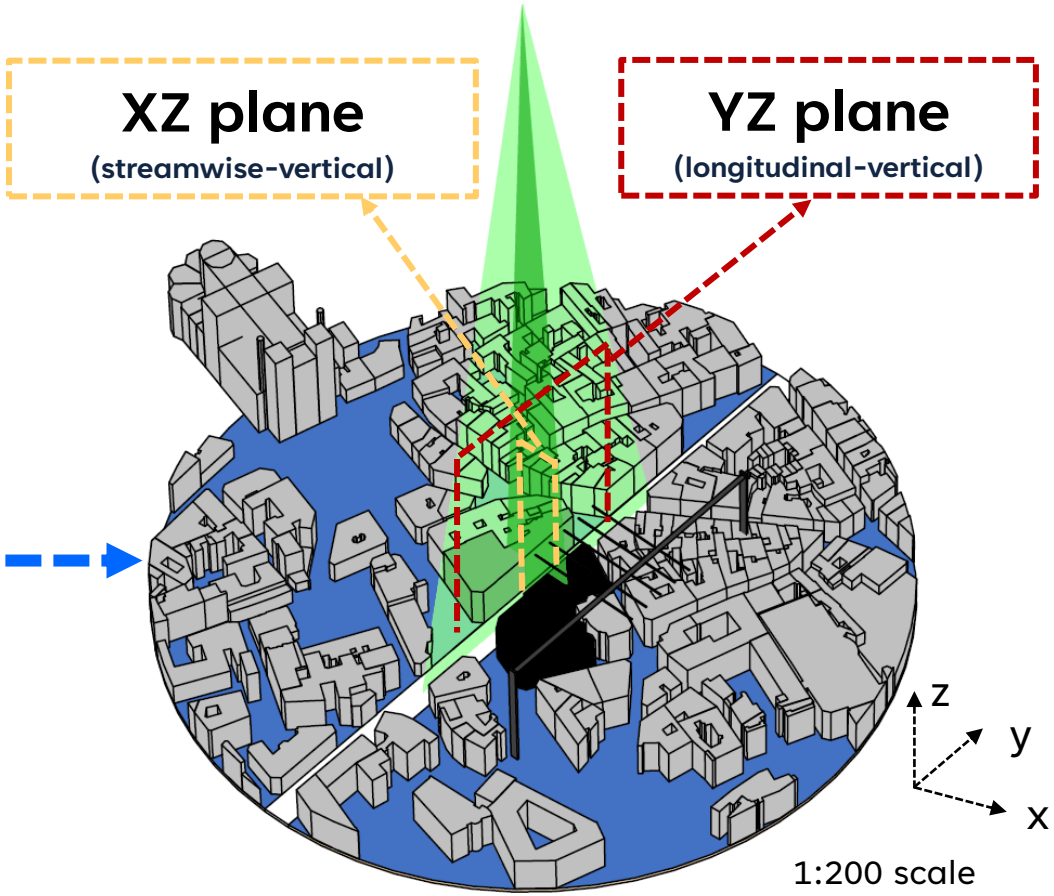
$h_1 = 50\ mm$ – cubic roughness height

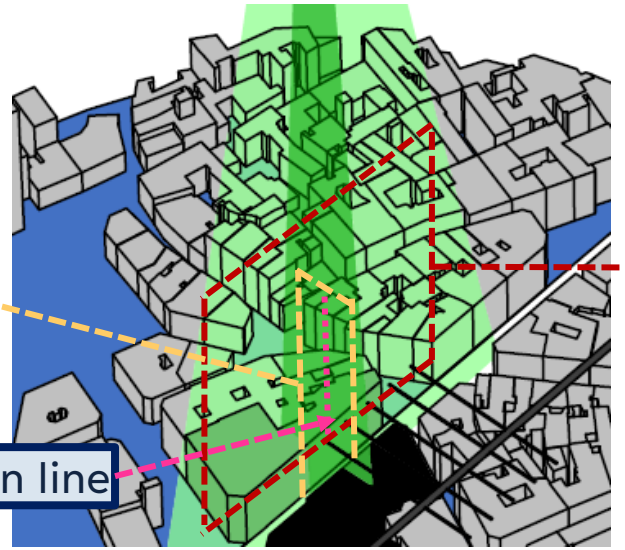
$h_2 = 116\ mm$ – street canyon ridge height

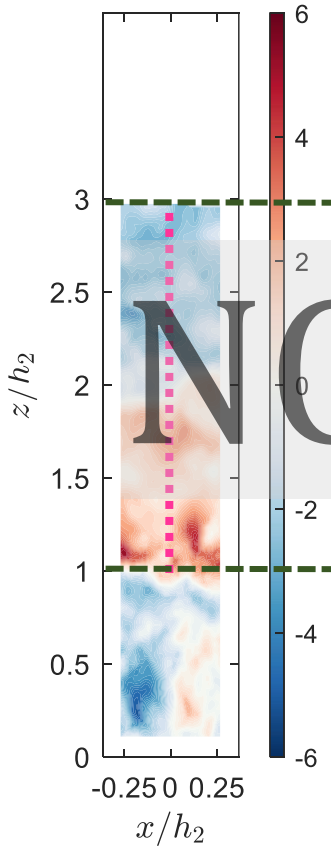
$\bar{h} = 100\ mm$ – RdS average height without the cathedral

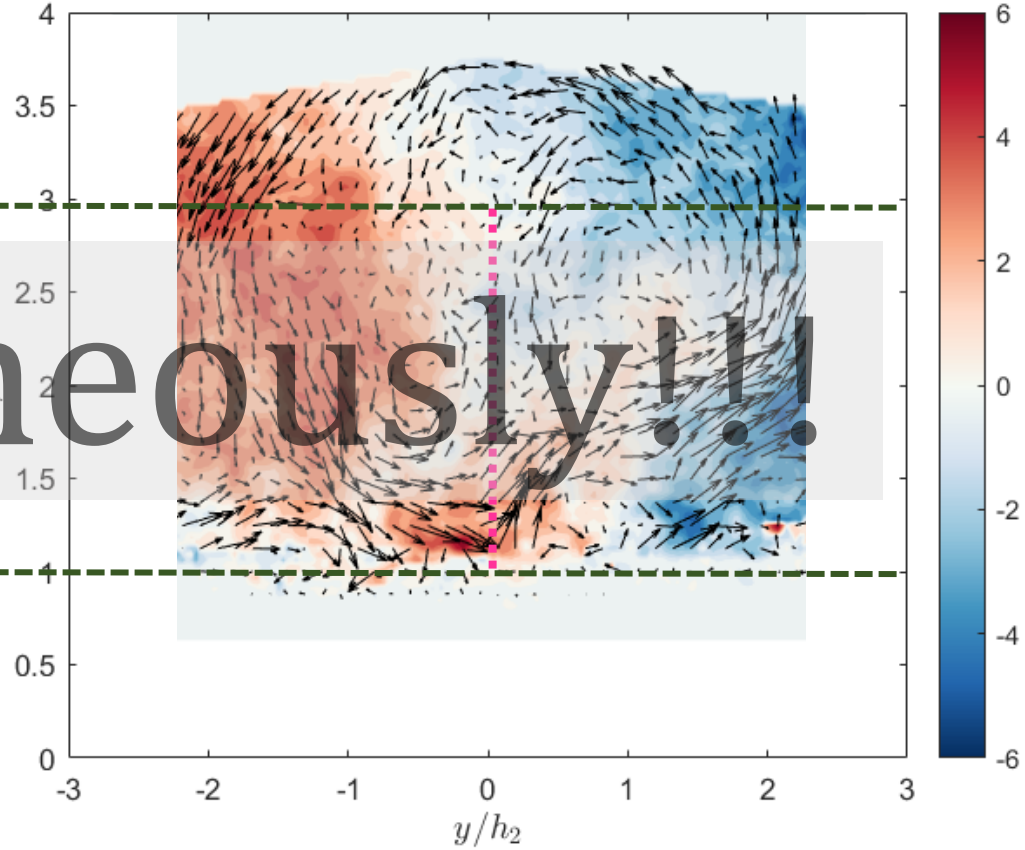












NOT simultaneously!!!

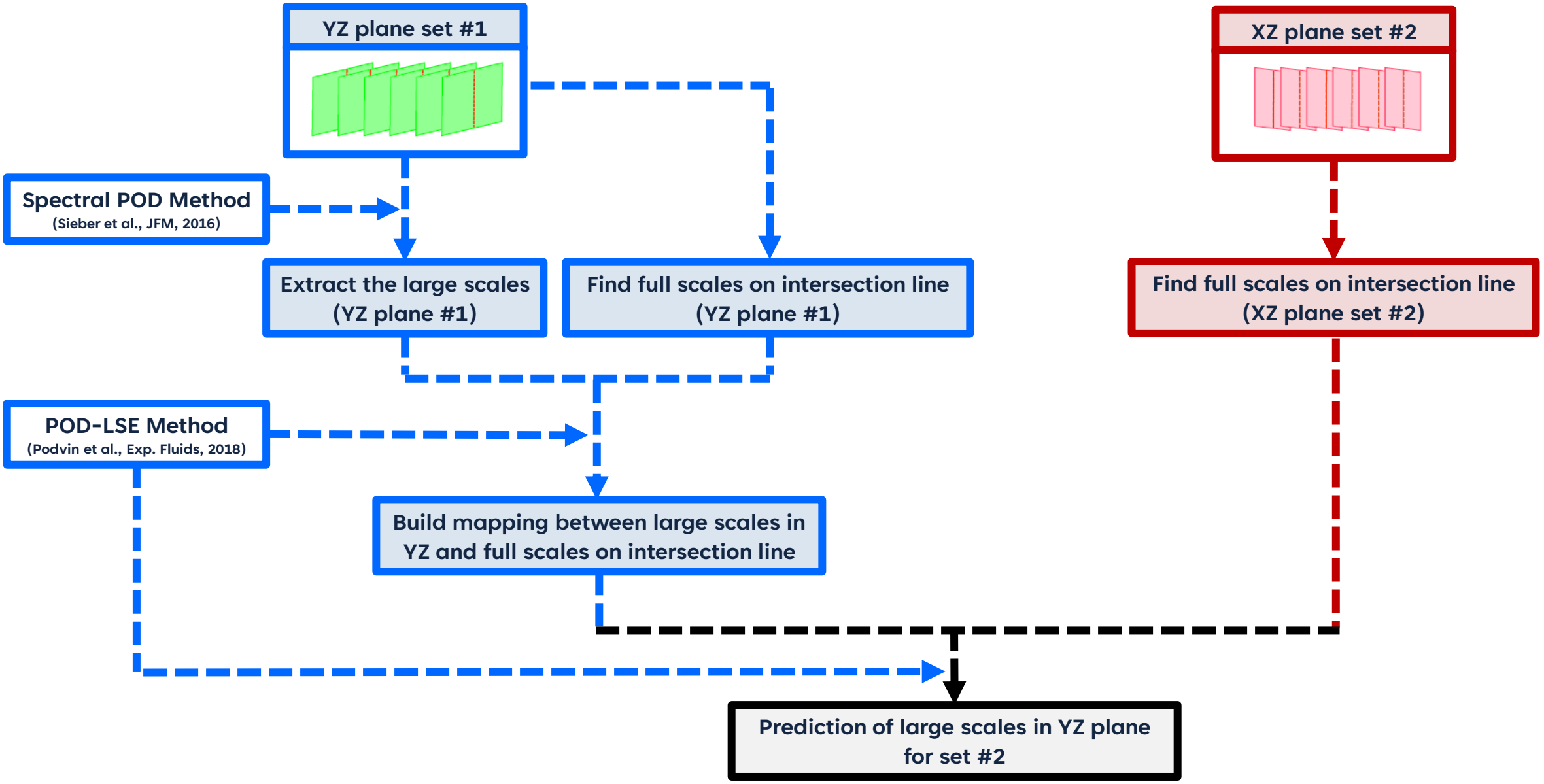
Limitations

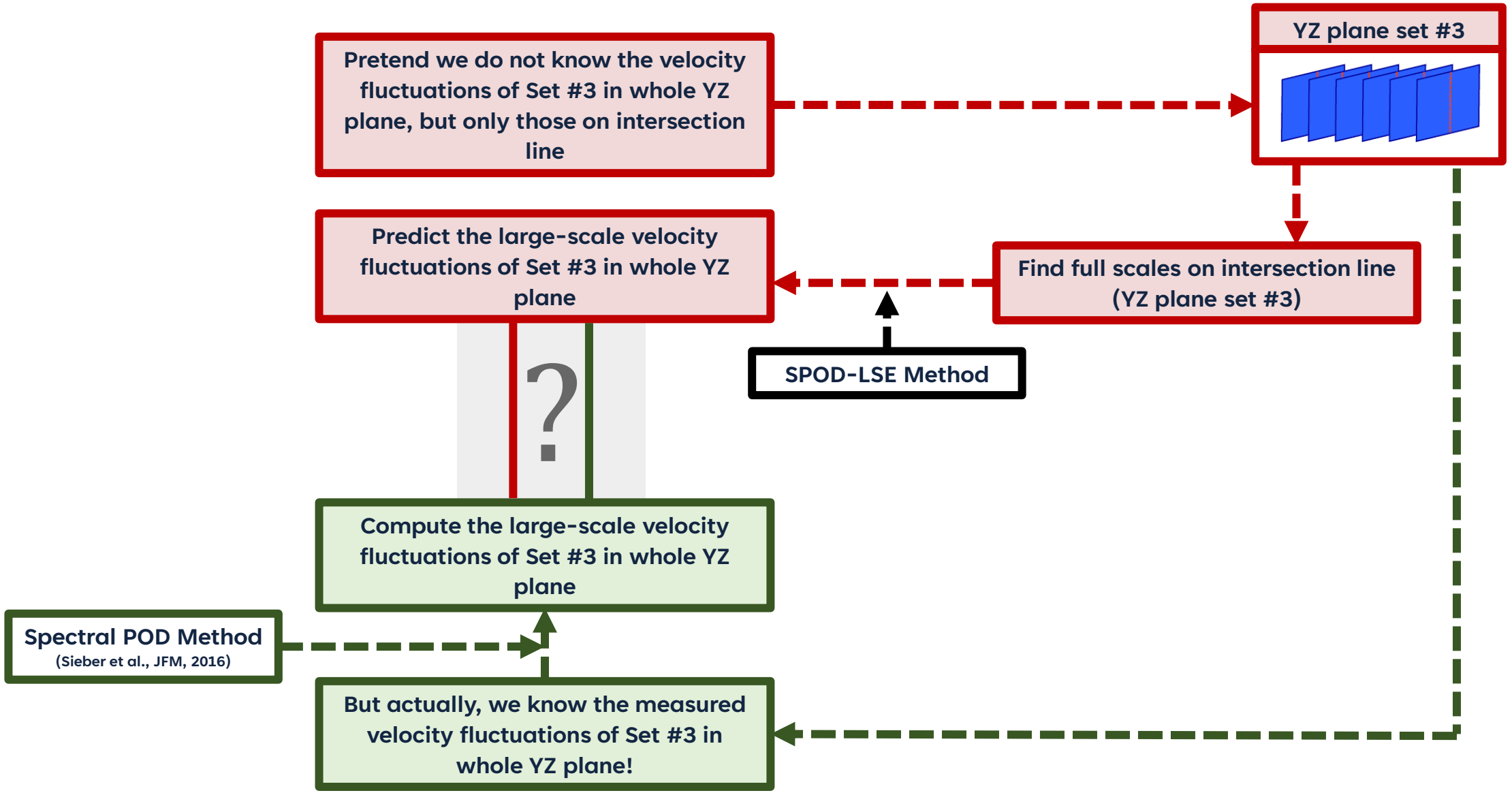
1. For Set #1 & #3, we have no information within the street canyon.
2. For Set #2, we have no information of the boundary layer flow in the YZ plane.

Challenges

1. Extraction of large scales in the boundary layer.
2. Synchronization of the YZ plane and XZ plane measurements.
3. Computation of correlations of some variables in boundary layer and street canyon.

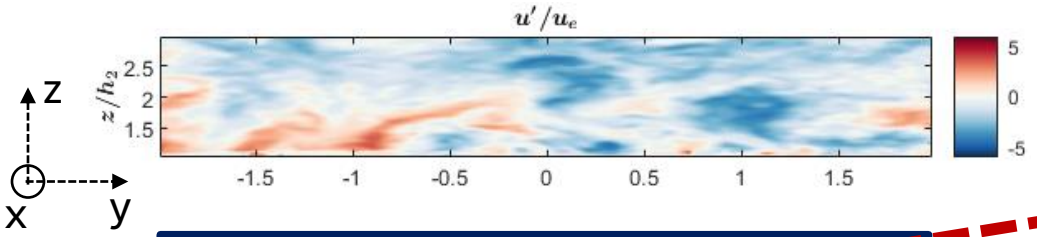
Research Techniques


YZ plane set #1
XZ plane set #2
Spectral POD Method
(Sieber et al., JFM, 2016)
Extract the large scales
(YZ plane #1)
Find full scales on intersection line
(YZ plane #1)
Find full scales on intersection line
(XZ plane set #2)
POD-LSE Method
(Podvin et al., Exp. Fluids, 2018)
Build mapping between large scales in
YZ and full scales on intersection line
Prediction of large scales in YZ plane
for set #2

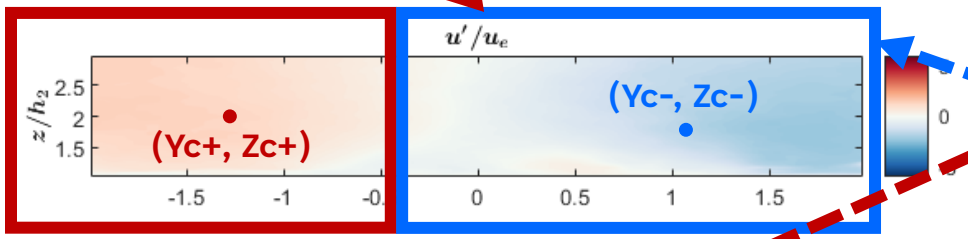

Pretend we do not know the velocity fluctuations of Set #3 in whole YZ plane, but only those on intersection line
YZ plane set #3
Predict the large-scale velocity fluctuations of Set #3 in whole YZ plane
Find full scales on intersection line (YZ plane set #3)
SPOD-LSE Method
?
Compute the large-scale velocity fluctuations of Set #3 in whole YZ plane
Spectral POD Method
(Sieber et al., JFM, 2016)
But actually, we know the measured velocity fluctuations of Set #3 in whole YZ plane!

Results

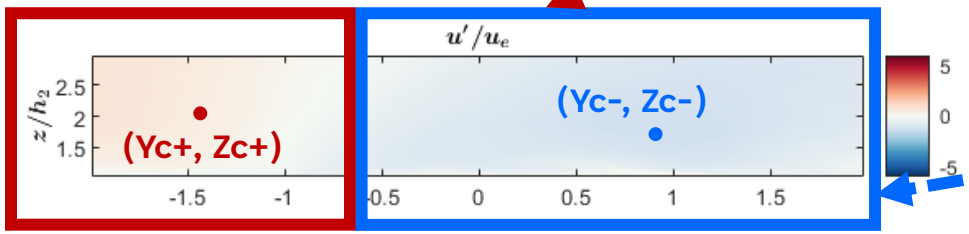
Snapshot #17777 from Set #3



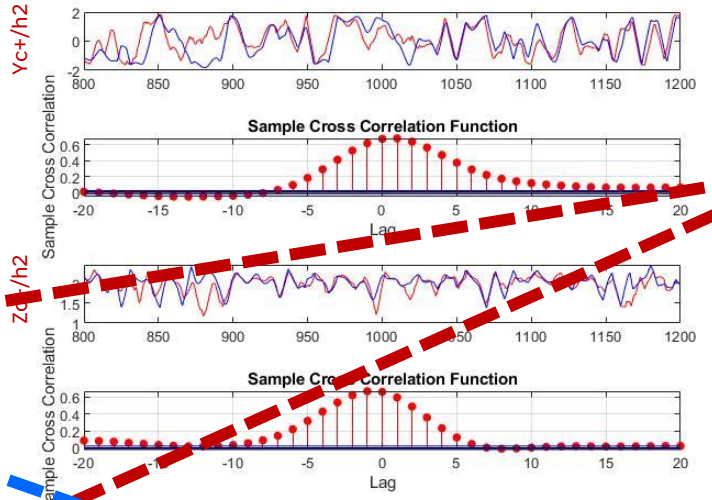
Large scales of #17777



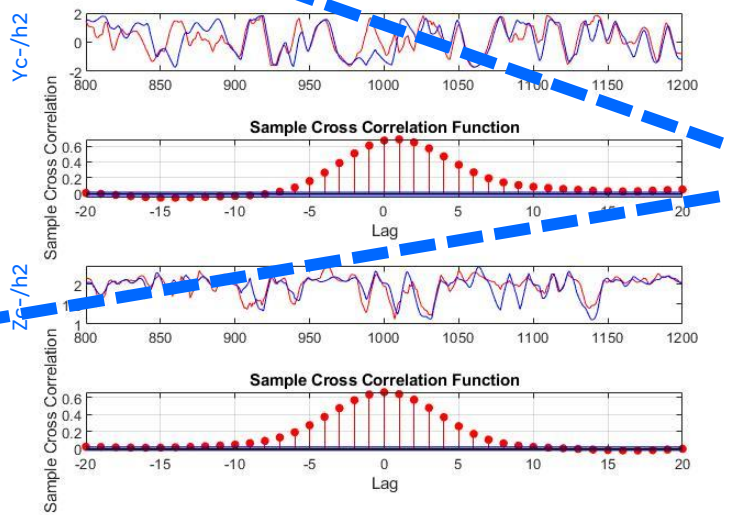
Prediction of #17777



Positive region



negative region



Find +/- fluctuations centres by energy

$$\begin{pmatrix} y_c^+(t) \\ z_c^+(t) \end{pmatrix} := \frac{\int u'^{+2}(y,z,t) \begin{pmatrix} y \\ z \end{pmatrix} dydz}{\int u'^{+2}(y,z,t) dydz}$$

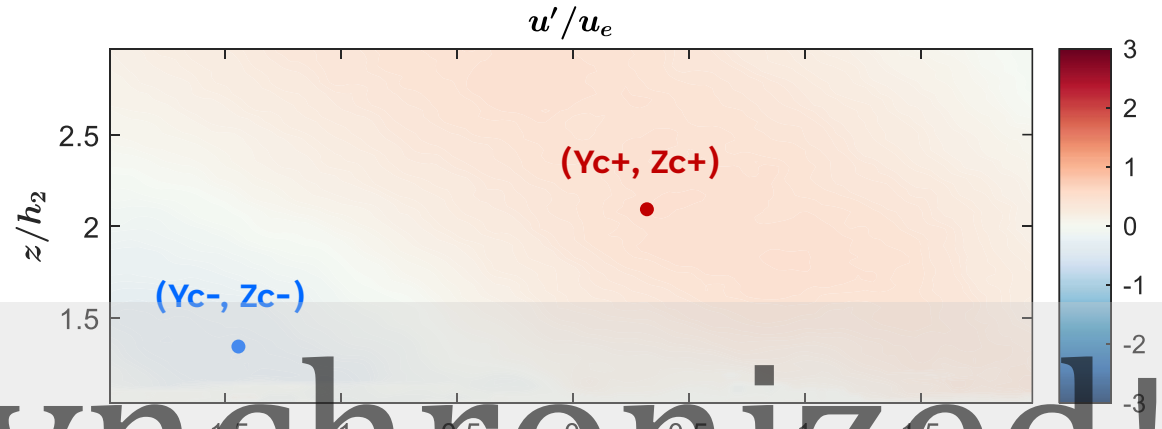
$$u'^{+}(y,z,t) = \begin{cases} u'(y,z,t), \text{if } u'(y,z,t) > 0 \\ 0, \text{if } u'(y,z,t) \leqslant 0 \end{cases}$$

$$\begin{pmatrix} y_c^-(t) \\ z_c^-(t) \end{pmatrix} := \frac{\int u'^{-2}(y,z,t) \begin{pmatrix} y \\ z \end{pmatrix} dydz}{\int u'^{-2}(y,z,t) dydz}$$

$$u'^{-}(y,z,t) = \begin{cases} u'(y,z,t), \text{if } u'(y,z,t) < 0 \\ 0, \text{if } u'(y,z,t) \geqslant 0 \end{cases}$$

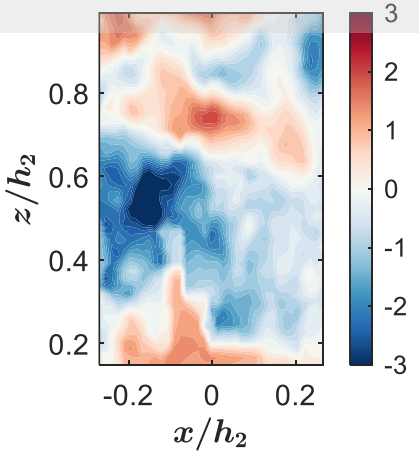
Example

YZ plane
Large-scale
Prediction

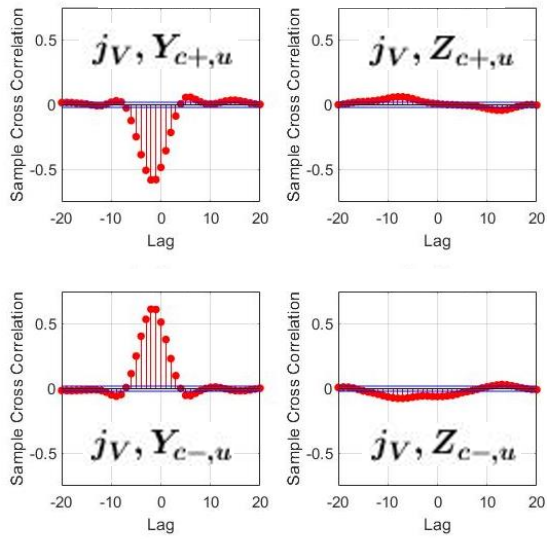
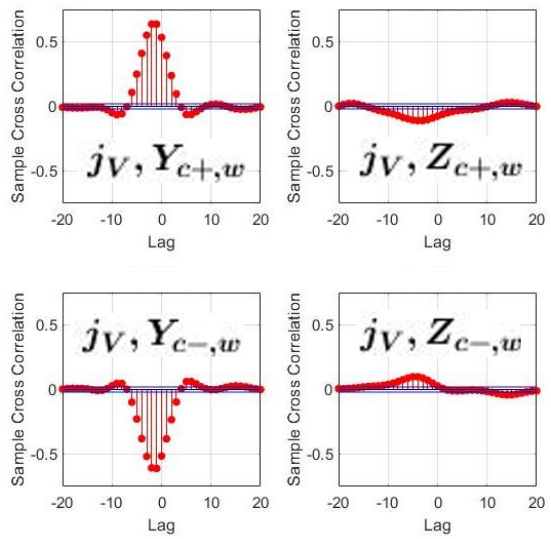
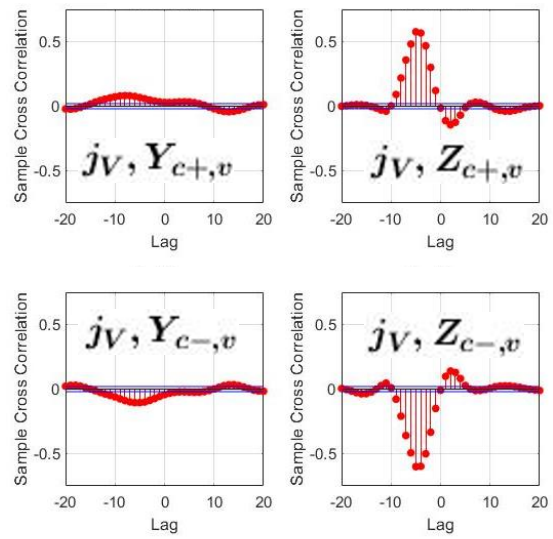
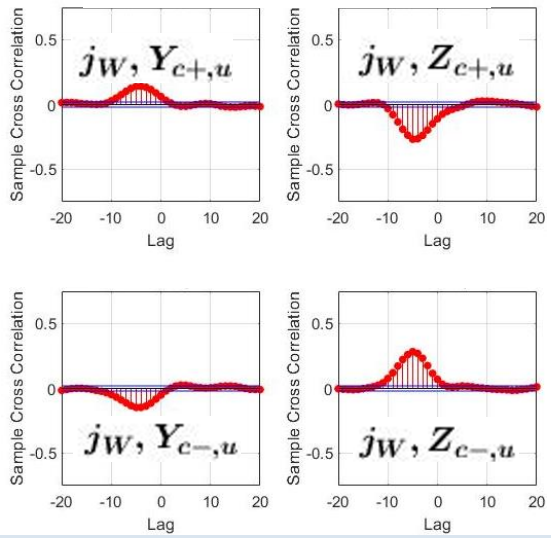
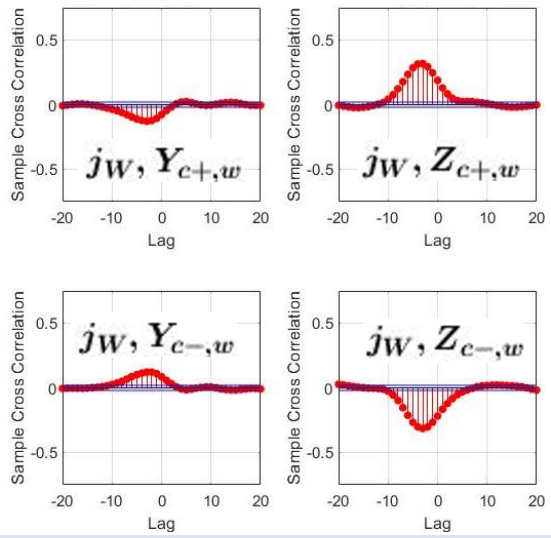
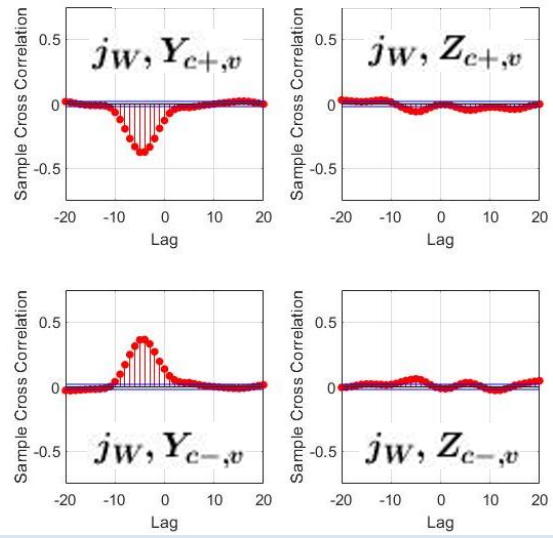


Synchronized!!!

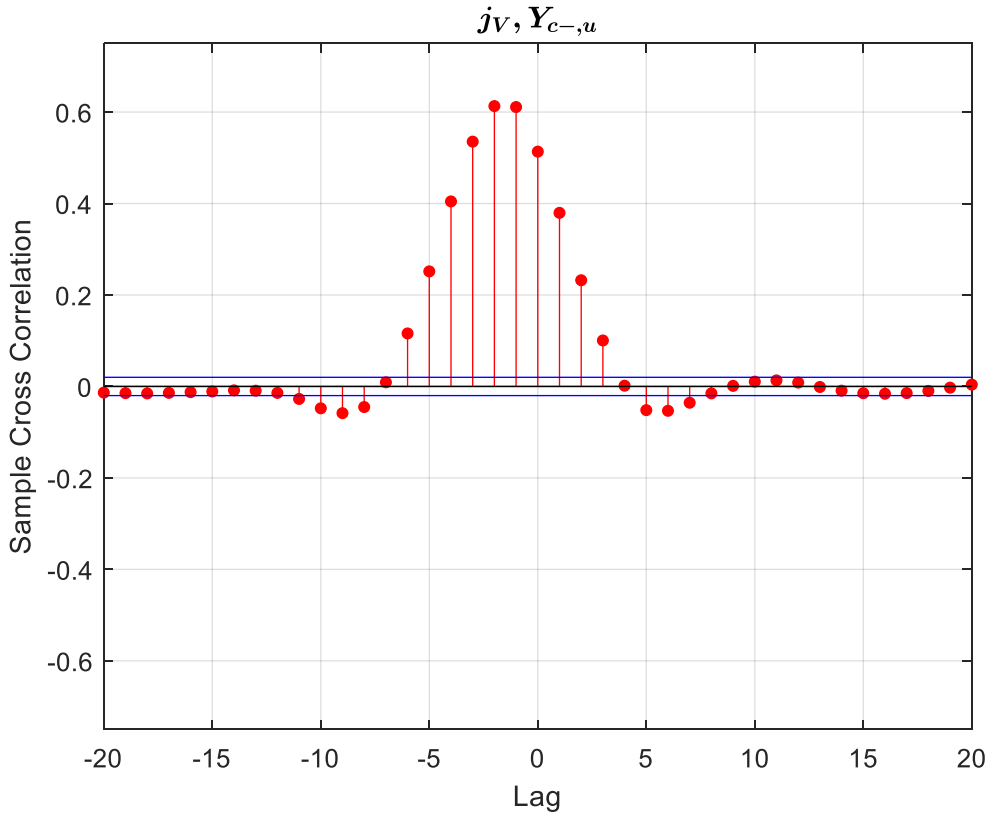
XZ plane
Full-scale
Measurement

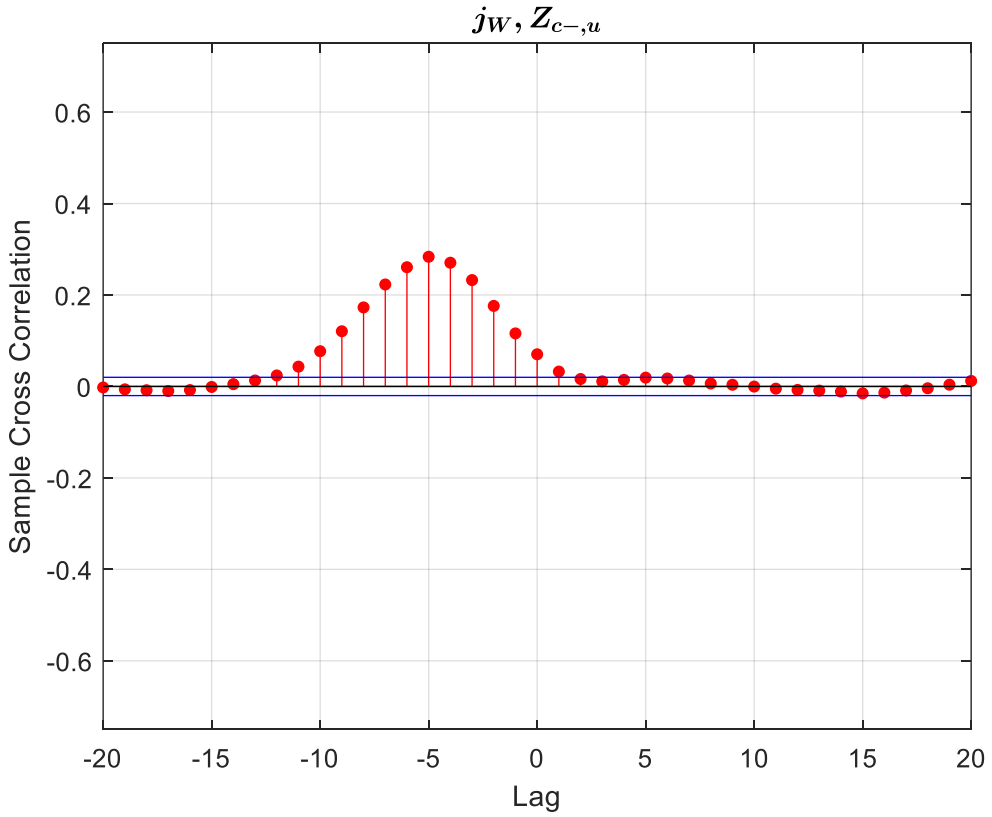


> Results > Prediction set #2 > Cross-correlation

	u’	w’	v’
$j_{V,i} = \frac{1}{U_e W_c h_2} \int_0^{h_2} \int_{-W_c/2}^{W_c/2} \rho_a V(x,z,i)\ dxdz$			
$j_{W,i} = \frac{1}{U_e W_c} \int_{-W_c/2}^{W_c/2} \rho_a W(x, z = h_2, i)\ dx$			

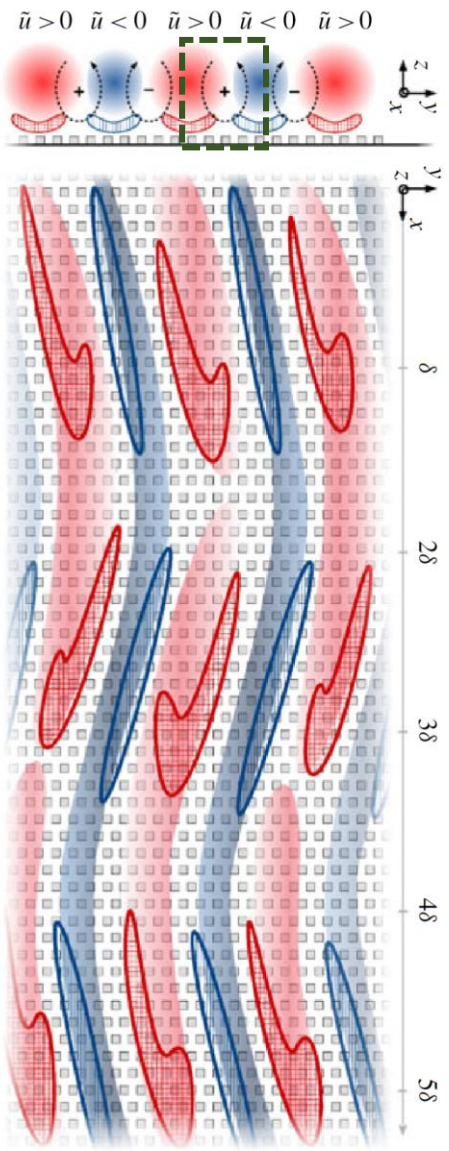
> Results > Prediction set #2 > Cross-correlation



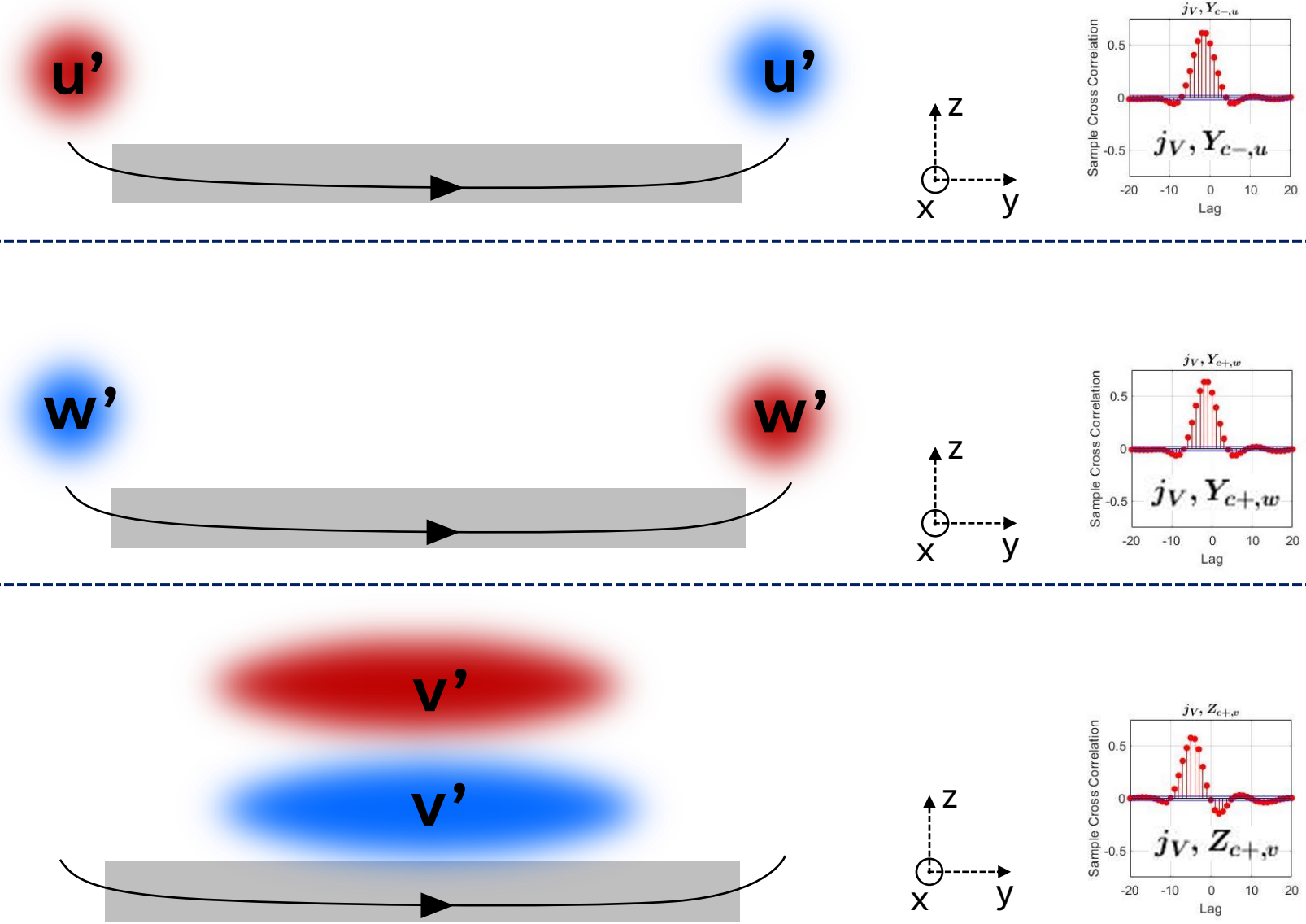


Conclusion

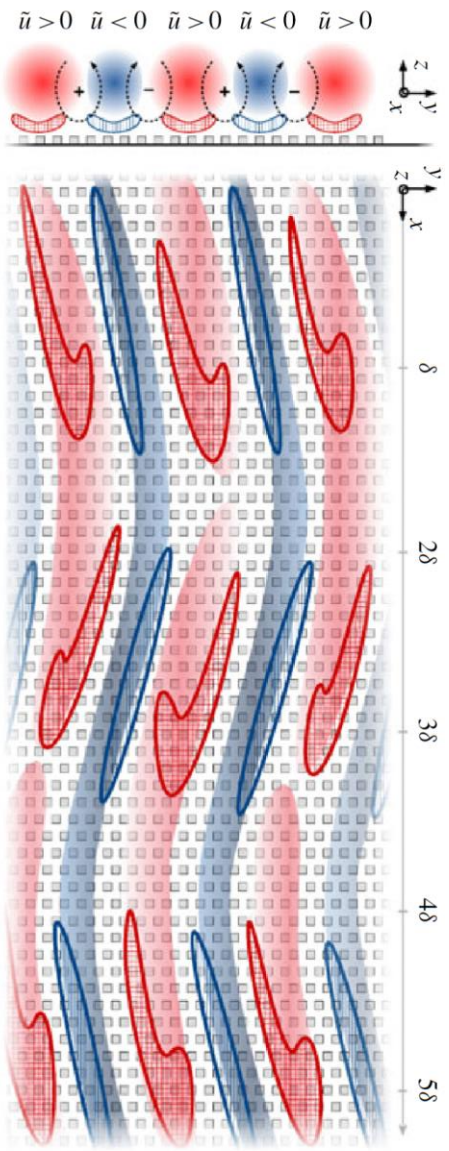
> Conclusions > Along-canyon mass flux (jv)



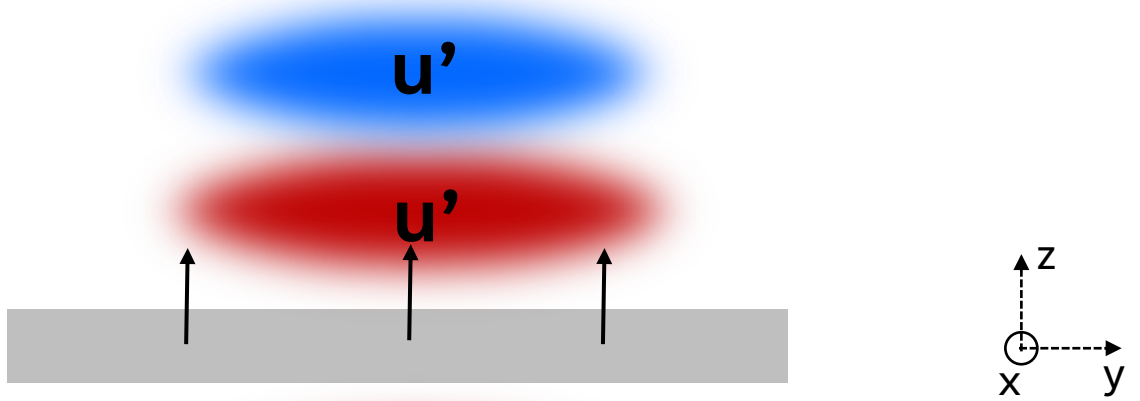
(Basley et al., JFM, 2016)

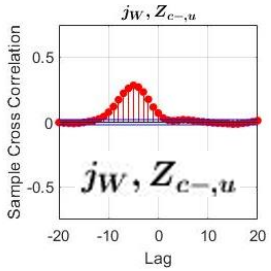


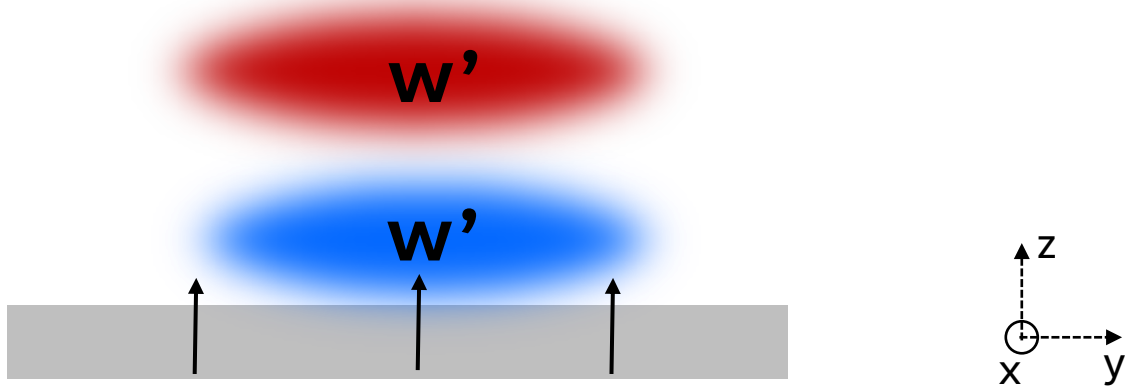
> Conclusions > Wall-normal mass flux (jw)

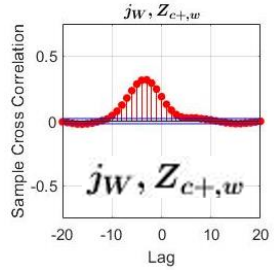


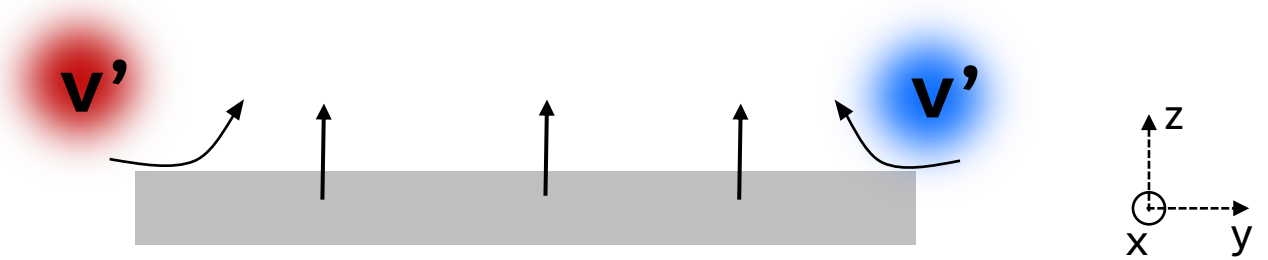
(Basley et al., JFM, 2016)

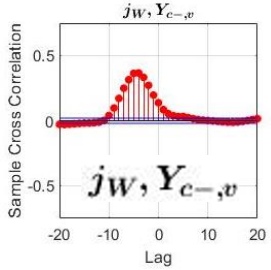












Summary

> Summary

1. We are able to predict the large-scales in the boundary layer by the same centreline above the street canyon.
2. The locations of the large-scale fluctuation centres have strong correlation with the mass flux in the along-canyon direction, while they also have decent correlation with the mass flux across roof-level.

Future Work

> Future Work

1. More analysis on correlation of the velocity fluctuation centre locations and mass flux in street canyon, for better understanding in physics.
2. Explore the influence of upstream tall building wake on the mass flux in street canyon

Thank you!

NSERC
CRSNG

Mitacs

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ANR

Western
UNIVERSITY · CANADA



Dr. Eric Savory

CENTRALE
NANTES



Dr. Laurent Perret

Spectral POD Method

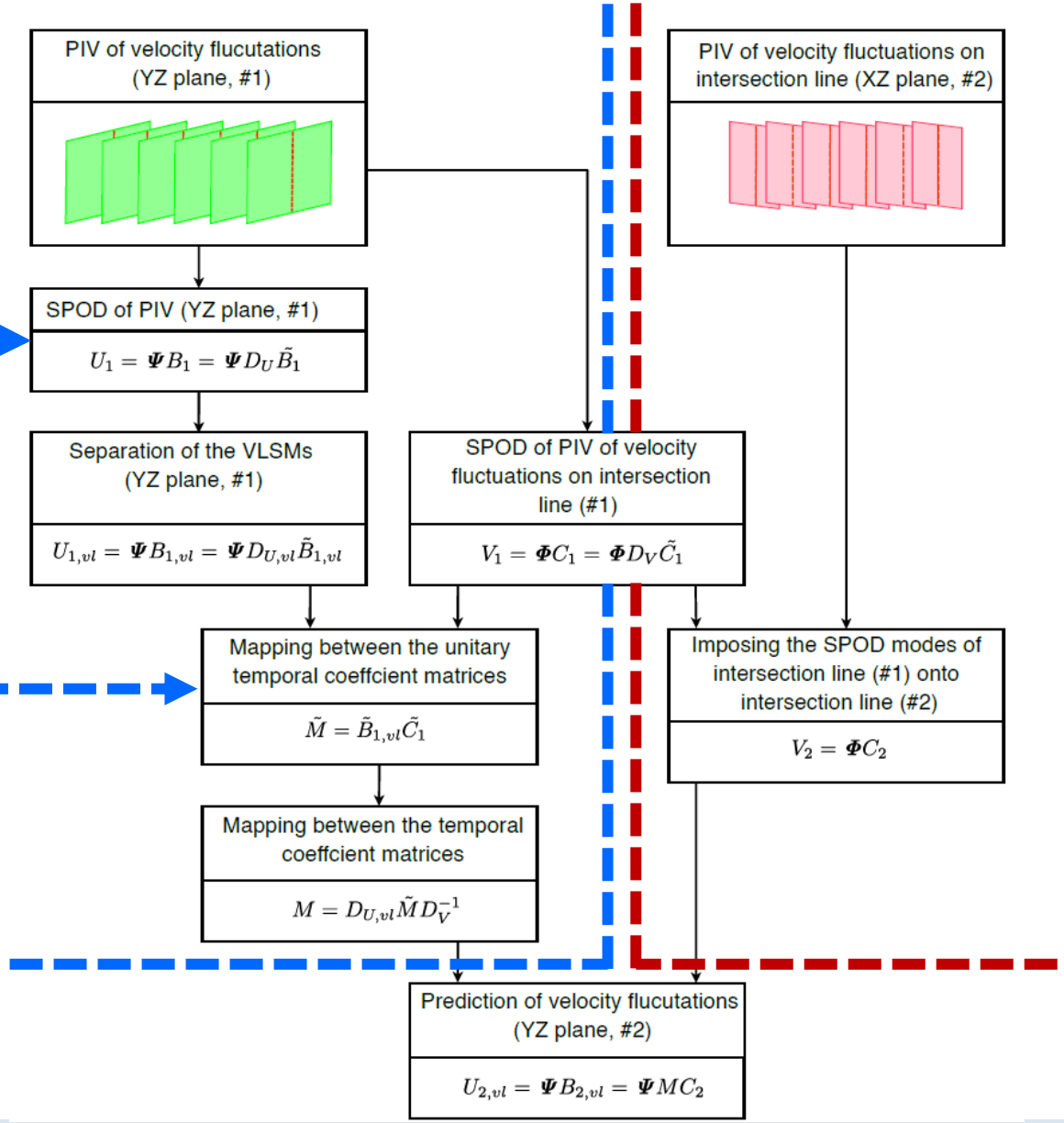
(Sieber et al., JFM, 2016)

Extract the large scales

POD-LSE Method

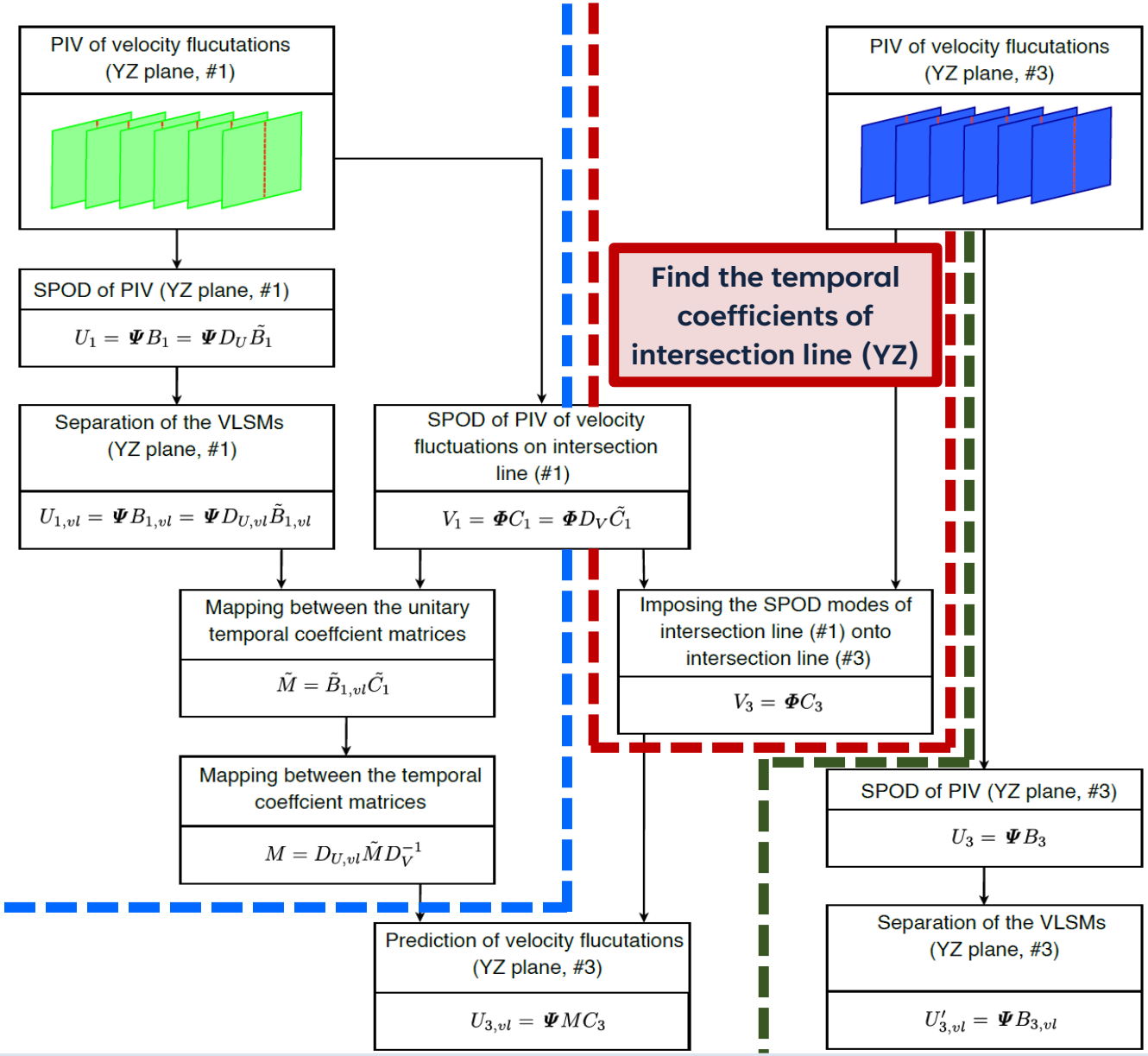
(Podvin et al., Exp. Fluids, 2018)

Find the relationship between the large scales in YZ plane and intersection line (YZ), by mapping the temporal coefficient matrix



Find the temporal coefficients of intersection line (XZ), since the SPOD spatial modes can be regarded as same as under same lab conditions

Symbols	Matrix of
U	Velocity fluctuations (YZ)
V	Velocity fluctuations (XZ)
Ψ, Φ	Spatial modes
B, C	Temporal coefficients
D_U, D_V	Sqrt of eigen values
$\tilde{B}, \tilde{C}$	Unitary temporal coefficients



Pretend we do not know the velocity fluctuations of Set #3 in whole YZ plane, but only those on intersection line

Predict the large-scale velocity fluctuations of Set #3 in whole YZ plane

?

Compute the large-scale velocity fluctuations of Set #3 in whole YZ plane

But actually, we know the measured velocity fluctuations of Set #3 in whole YZ plane!

Symbols	Matrix of
U	Velocity fluctuations (YZ)
V	Velocity fluctuations (XZ)
Ψ, Φ	Spatial modes
B, C	Temporal coefficients
D_U, D_V	Sqrt of eigen values
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